

Worksheet

Do the following series converge or diverge? State the test you used. For convergent series, indicate if they converge conditionally or absolutely. If appropriate, find the sum.

1. $\sum_{k=1}^{\infty} \frac{\cos k}{k^2}$

2. $\sum_{k=1}^{\infty} \frac{k-1}{k^{3/2}}$

3. $\sum_{k=1}^{\infty} k \sin\left(\frac{1}{k}\right)$

4. $\sum_{k=1}^{\infty} \frac{(-1)^k 7^{2k}}{(2k)!}$

5. Can the ratio test be used for the series $\sum_{k=0}^{\infty} \frac{(-1)^k k^2}{3^k}$? Completely explain your answer.